LAB 7

Fuzzing for Input Validation Bugs

Student Grades Application

Name: DEEKSHITH GOWDA G

SRN: PES1UG24CS142

Objective

The objective of this lab is to use the Hypothesis fuzzing library to test the student grades Python application, identify input-validation bugs, fix the bugs, and run the tests again.

Tools Used

Python 3.11.9, pytest, and the Hypothesis library.

Functions Tested

The three functions tested are average_marks(), parse_srn(), and short_name(). Hypothesis generates different text values and None inputs to check whether the functions crash.

1. Original Application Execution

The original application was executed with valid sample inputs. It calculated the average marks as 85.0, extracted the campus, year and department from the SRN, and displayed the shortened name as D. Gowda.

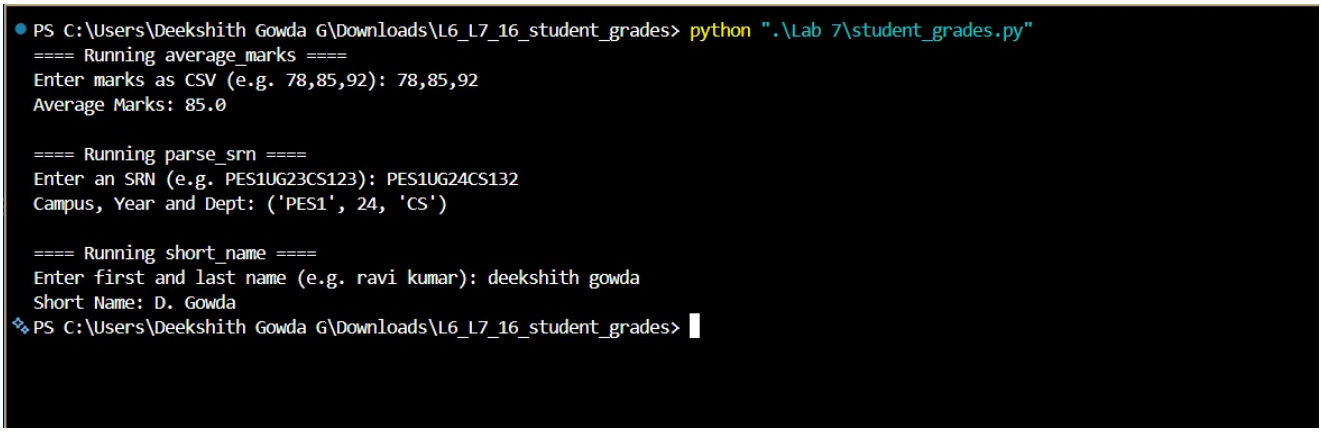


Figure 1: Original application output.

2. Initial Fuzz Testing

The initial Hypothesis tests found input-validation problems in all three functions. Empty strings and None values caused failures, and all three tests were reported as failed.

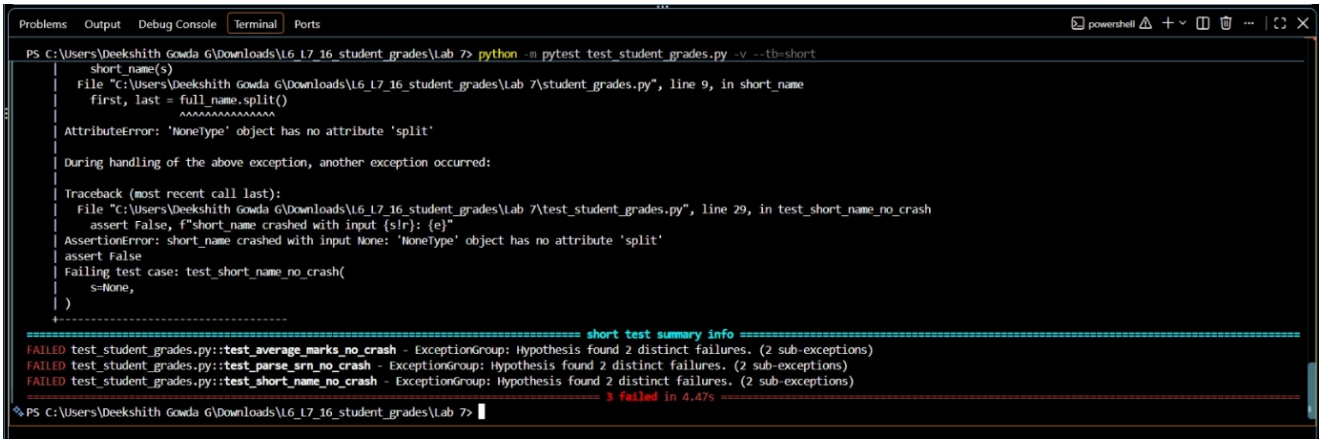


Figure 2: Initial fuzz-testing failures.

3. Fixed Application Execution

The application was updated with input validation and error handling. It was executed again using valid inputs, and all three functions produced the expected output.

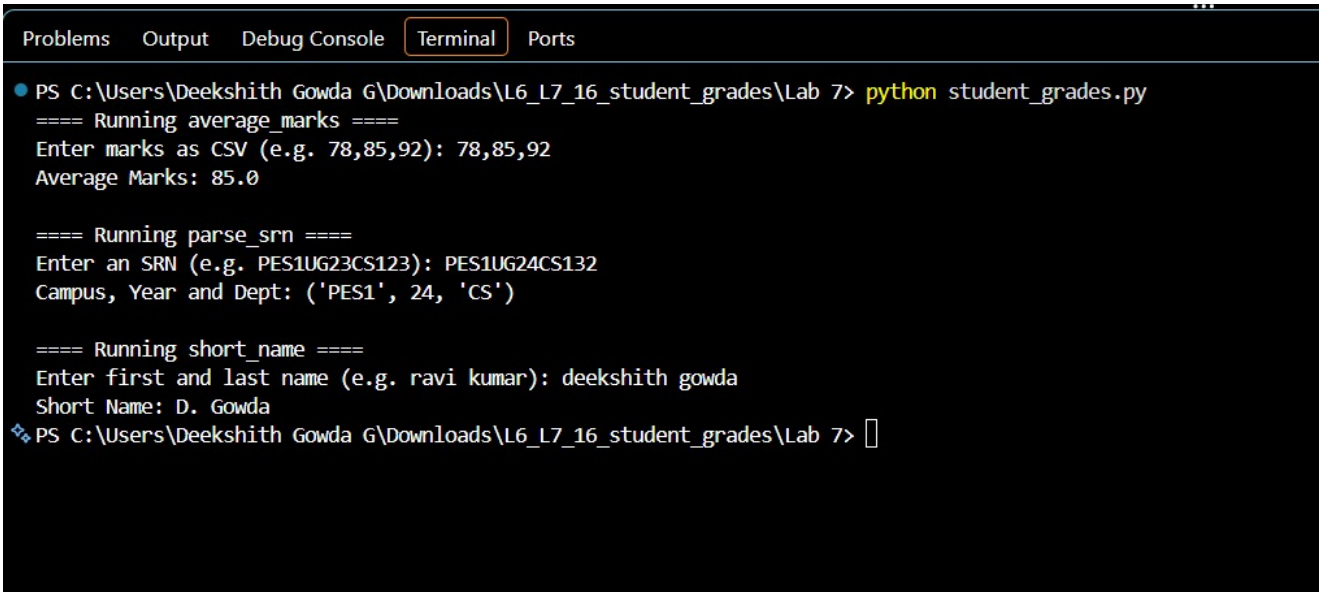


Figure 3: Fixed application output.

4. Final Fuzz Testing

After the fixes, the Hypothesis tests were executed again. All three tests passed successfully: average_marks(), parse_srn(), and short_name(). The final pytest result was 3 passed.

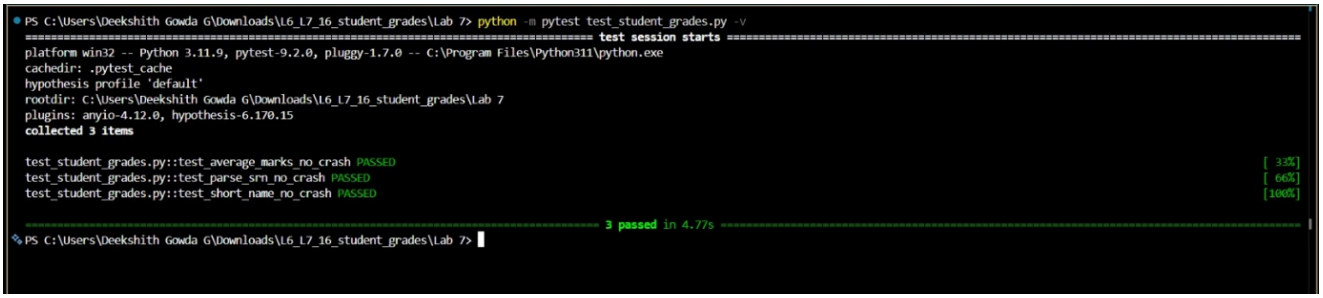


Figure 4: Final test result showing 3 passed.

Conclusion

In this lab, Hypothesis was used to identify input-validation bugs in the student grades application. The three functions were updated to handle the tested unexpected inputs. After fixing the code, all three fuzz tests passed. This demonstrates how fuzz testing helps improve the robustness of a Python application.